

WORKED EXAMPLE

An activation and its slope are different

Lecture 3, PDF page 156 lists the four functions and their derivatives.

$$z = b + \sum_i w_i x_i, \quad a = f(z), \quad f'(z) = \frac{da}{dz}$$

z is the weighted input before activation. We write a for the activation; the notes often write y . Here a is not a known target label.

The derivative is the local slope: how much the activation changes for a tiny increase in z .

$$\Delta a \approx f'(z) \Delta z$$

Our main input is $z = \ln(3)$. Natural logarithm $\ln(3)$ means the number whose exponential is 3. This choice makes hand arithmetic simple.

$$e^{\ln 3} = 3, \quad e^{-\ln 3} = 1/3, \quad \ln 3 \approx 1.098612$$

WORKED EXAMPLE

Sigmoid: calculate the activation first

The output is between zero and one; its derivative is a different number.

$$a = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$z = \ln 3: \quad a = \frac{1}{1 + 1/3} = \frac{1}{4/3} = \frac{3}{4} = 0.75$$

$$\sigma'(z) = a(1 - a)$$

$$\sigma'(\ln 3) = \frac{3}{4} \left(1 - \frac{3}{4}\right) = \frac{3}{4} \frac{1}{4} = \frac{3}{16} = 0.1875$$

$$\Delta z = 0.01 \Rightarrow \Delta a \approx 0.1875(0.01) = 0.001875$$

A small increase of 0.01 in input changes this activation by about 0.001875. This is a local approximation, not an exact finite difference.

WORKED EXAMPLE

Tanh: use the two exponential values

The assignment gives this exponential form; the lecture gives the derivative.

$$a = \tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$a = \frac{3 - 1/3}{3 + 1/3} = \frac{8/3}{10/3} = \frac{8}{10} = \frac{4}{5} = 0.8$$

$$\tanh'(z) = 1 - a^2$$

$$\tanh'(\ln 3) = 1 - (4/5)^2 = 1 - 16/25 = 9/25 = 0.36$$

Square the activation before subtracting it from 1. The formula is not $(1 - a)$ squared. Unlike sigmoid, tanh may output negative values.

WORKED EXAMPLE

ReLU: choose the correct branch

ReLU means rectified linear unit: keep positive input, otherwise output zero.

$$a = \max(0, z)$$

$$z = \ln 3 > 0 \Rightarrow a = \ln 3 \approx 1.098612$$

$$f'(z) = 1 \quad \text{for } z > 0$$

$$f'(\ln 3) = 1$$

At a positive input, increasing z by 0.01 increases the activation by exactly 0.01 as long as both inputs remain positive.

Zero is a special point where the classical derivative does not exist. We will compare the source conventions explicitly.

WORKED EXAMPLE

Softplus: a smooth positive activation

Use the natural logarithm, not a base-10 calculator logarithm.

$$a = \ln(1 + e^z)$$

$$a = \ln(1 + 3) = \ln 4 \approx 1.386294$$

$$f'(z) = \frac{1}{1 + e^{-z}} = \sigma(z)$$

$$f'(\ln 3) = \frac{1}{1 + 1/3} = \frac{3}{4} = 0.75$$

Softplus and sigmoid have different outputs. Softplus has sigmoid as its derivative; do not replace its activation by sigmoid.

WORKED EXAMPLE

Negative input: sigmoid and tanh

Now $z = -\ln(3)$, so $\exp(z) = 1/3$ and $\exp(-z) = 3$.

$$\sigma(-\ln 3) = \frac{1}{1+3} = \frac{1}{4} = 0.25$$

$$\sigma'(-\ln 3) = \frac{1}{4}(1 - 1/4) = \frac{3}{16} = 0.1875$$

$$\tanh(-\ln 3) = \frac{1/3 - 3}{1/3 + 3} = \frac{-8/3}{10/3} = -\frac{4}{5}$$

$$\tanh'(-\ln 3) = 1 - (-4/5)^2 = 1 - 16/25 = 9/25$$

The activation of tanh is negative, but its slope here is positive. A negative activation does not imply a negative derivative.

WORKED EXAMPLE

Negative input: ReLU and softplus

The two functions behave differently on the negative side.

$$\text{ReLU}(-\ln 3) = \max(0, -\ln 3) = 0$$

$$\text{ReLU}'(-\ln 3) = 0$$

$$\text{softplus}(-\ln 3) = \ln(1 + 1/3) = \ln(4/3) \approx 0.287682$$

$$\text{softplus}'(-\ln 3) = \frac{1}{1+3} = \frac{1}{4}$$

ReLU is flat for strictly negative input. Softplus is smooth and still has a small positive output and slope.

WORKED EXAMPLE

At zero: three smooth functions

Substitute $\exp(0) = 1$ before calculating the derivatives.

$$\sigma(0) = \frac{1}{1+1} = \frac{1}{2}, \quad \sigma'(0) = \frac{1}{2}(1 - 1/2) = \frac{1}{4}$$

$$\tanh(0) = \frac{1-1}{1+1} = 0, \quad \tanh'(0) = 1 - 0^2 = 1$$

$$\text{softplus}(0) = \ln(1 + 1) = \ln 2 \approx 0.693147$$

$$\text{softplus}'(0) = \frac{1}{1+1} = \frac{1}{2}$$

Zero activation and zero derivative are different claims. Tanh has activation 0 at the origin but slope 1.

WORKED EXAMPLE

ReLU at zero: reconcile the two sources

The left and right slopes disagree, so there is no classical derivative.

$$\text{ReLU}(0) = 0$$

$$\text{left slope} = \frac{f(0) - f(-0.01)}{0.01} = \frac{0 - 0}{0.01} = 0$$

$$\text{right slope} = \frac{f(0.01) - f(0)}{0.01} = \frac{0.01 - 0}{0.01} = 1$$

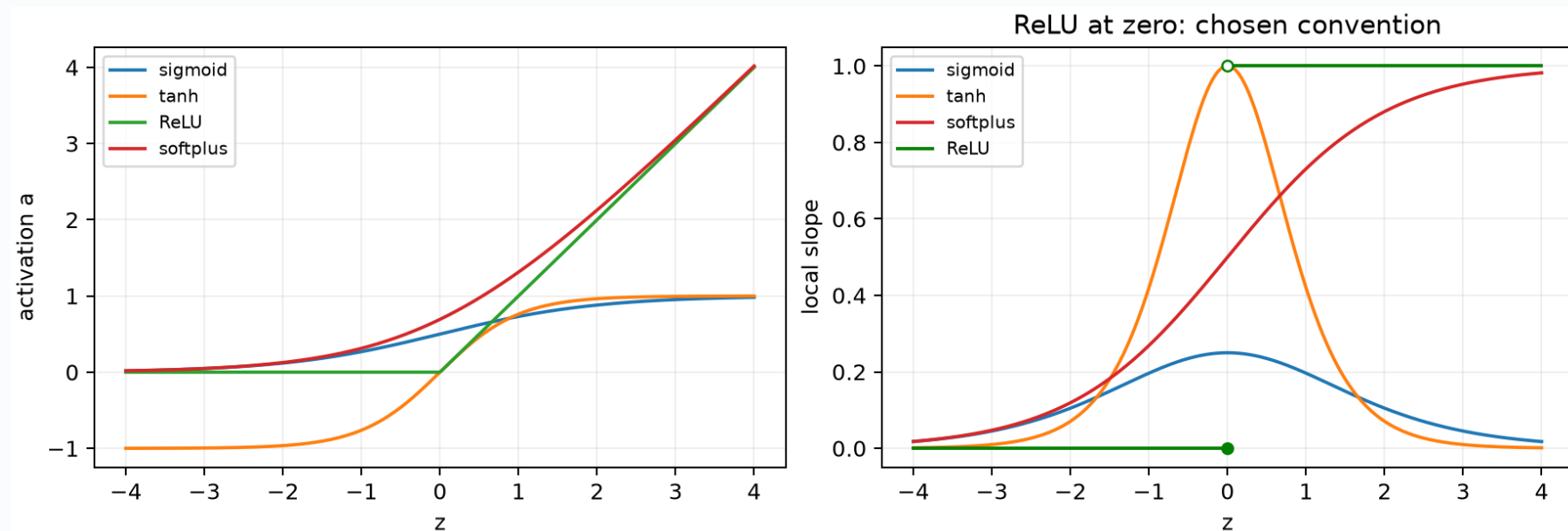
Lecture 3, PDF page 156 assigns slope 1 when $z \geq 0$. Assignment 1, PDF page 2 assigns slope 1 only when $z > 0$, otherwise 0.

These are chosen backpropagation conventions at the kink, not classical derivatives. For assignment-aligned work, use 0 at zero and state that convention.

WORKED EXAMPLE

Compare the activation curves

Same input z , different outputs and local slopes.



Sigmoid and tanh flatten near their upper and lower limits. ReLU has a corner at zero. Softplus smoothly bends from a near-flat negative side to a near-linear positive side.

WORKED EXAMPLE

Saturation means a small local slope

At $z = 4$ the smooth bounded activations change slowly; their slopes are not zero.

$$\sigma(4) = \frac{1}{1 + e^{-4}} \approx 0.982014$$

$$\sigma'(4) = 0.982014(1 - 0.982014) \approx 0.017663$$

$$\tanh(4) \approx 0.999329, \quad \tanh'(4) = 1 - 0.999329^2 \approx 0.001341$$

$$\Delta z = 0.01 : \quad \Delta \sigma \approx 0.00017663, \quad \Delta \tanh \approx 0.00001341$$

At zero the same small input change gives about 0.0025 for sigmoid and 0.01 for tanh. Repeated small slope factors can shrink gradients in a deep network.

WORKED EXAMPLE

Use the local slope in the chain rule

Suppose the incoming loss derivative is $dL/da = 2$ at $z = \ln(3)$.

$$\frac{dL}{dz} = \frac{dL}{da} \frac{da}{dz} = 2f'(z)$$

sigmoid : $2(3/16) = 3/8 = 0.375$

tanh : $2(9/25) = 18/25 = 0.72$

ReLU : $2(1) = 2$

softplus : $2(3/4) = 3/2 = 1.5$

The incoming derivative depends on the rest of the network and loss. Here it is a supplied value, used to isolate the role of the activation derivative.

INDEPENDENT PRACTICE

Your turn: an input with simple exponentials

Use $z = \ln(2)$, so $\exp(z) = 2$ and $\exp(-z) = 1/2$.

For sigmoid, tanh, ReLU and softplus, calculate both the activation and its local derivative. Show the substitutions before simplifying.

Then suppose $dL/da = -3$. Calculate dL/dz for each activation using the chain rule.

Finally, state the derivative convention you would use for ReLU at zero in Assignment 1, and why it is a convention.

WORKED ANSWER

Practice answer: sigmoid and tanh

Keep exact fractions until the final result.

$$\sigma(\ln 2) = \frac{1}{1 + 1/2} = \frac{2}{3}$$

$$\sigma'(\ln 2) = \frac{2}{3}(1 - 2/3) = \frac{2}{9}$$

$$\frac{dL}{dz} = (-3)(2/9) = -2/3$$

$$\tanh(\ln 2) = \frac{2 - 1/2}{2 + 1/2} = \frac{3/2}{5/2} = \frac{3}{5}$$

$$\tanh'(\ln 2) = 1 - (3/5)^2 = 16/25$$

$$\frac{dL}{dz} = (-3)(16/25) = -48/25 = -1.92$$

WORKED ANSWER

Practice answer: ReLU and softplus

The sign of the loss derivative comes from the incoming value -3.

$$\text{ReLU}(\ln 2) = \ln 2 \approx 0.693147, \quad f' = 1$$

$$\frac{dL}{dz} = (-3)(1) = -3$$

$$\text{softplus}(\ln 2) = \ln(1 + 2) = \ln 3 \approx 1.098612$$

$$f' = \frac{1}{1 + 1/2} = \frac{2}{3}, \quad \frac{dL}{dz} = (-3)(2/3) = -2$$

Assignment 1 uses ReLU slope 0 at zero. Its left and right slopes are 0 and 1, so the usual derivative does not exist there.

Checklist: do not confuse activation with slope, tanh derivative with $(1-a)^2$, or softplus output with sigmoid output.